

Nutrition

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Symptoms of Metastatic Cancer

- Metastatic cancer **does not always cause symptoms.**
- When symptoms do occur, what they are like and how often you have them will depend on the size and location of the metastatic tumors.

Some common signs of metastatic cancer include:

- Pain and fractures, when cancer has spread to the bone
- Headache, seizures, or dizziness, when cancer has spread to the brain
- Shortness of breath, when cancer has spread to the lung
- Jaundice or swelling in the belly, when cancer has spread to the liver

Treatment for Metastatic Cancer

- There are treatments for most types of metastatic cancer.
- Often, the goal of treating metastatic cancer is to control it by stopping or slowing its growth.
- Some people can live for years with metastatic cancer that is well controlled. Other treatments may improve the [quality of life](#) by relieving symptoms.

Ongoing Research

Researchers are studying new ways to kill or stop the growth of primary and metastatic cancer cells. These ways include:

- Helping your [immune system](#) fight cancer
- Disrupting the steps in the process that allow the cancer cells to spread
- Targeting specific [genetic](#) changes in tumors

Carcinoma:

Cancer that begins in the skin or in tissues that line or cover internal organs.

Sarcoma:

A type of cancer that begins in bone or in the soft tissues of the body, including cartilage, fat, muscle, blood vessels, fibrous tissue, or other connective or supportive tissue.

Leukemia:

Cancer that starts in blood-forming tissue, such as the bone marrow, and causes large numbers of abnormal blood cells to be produced and enter the bloodstream.

The differences between normal cells and cancer cells

Normal cells

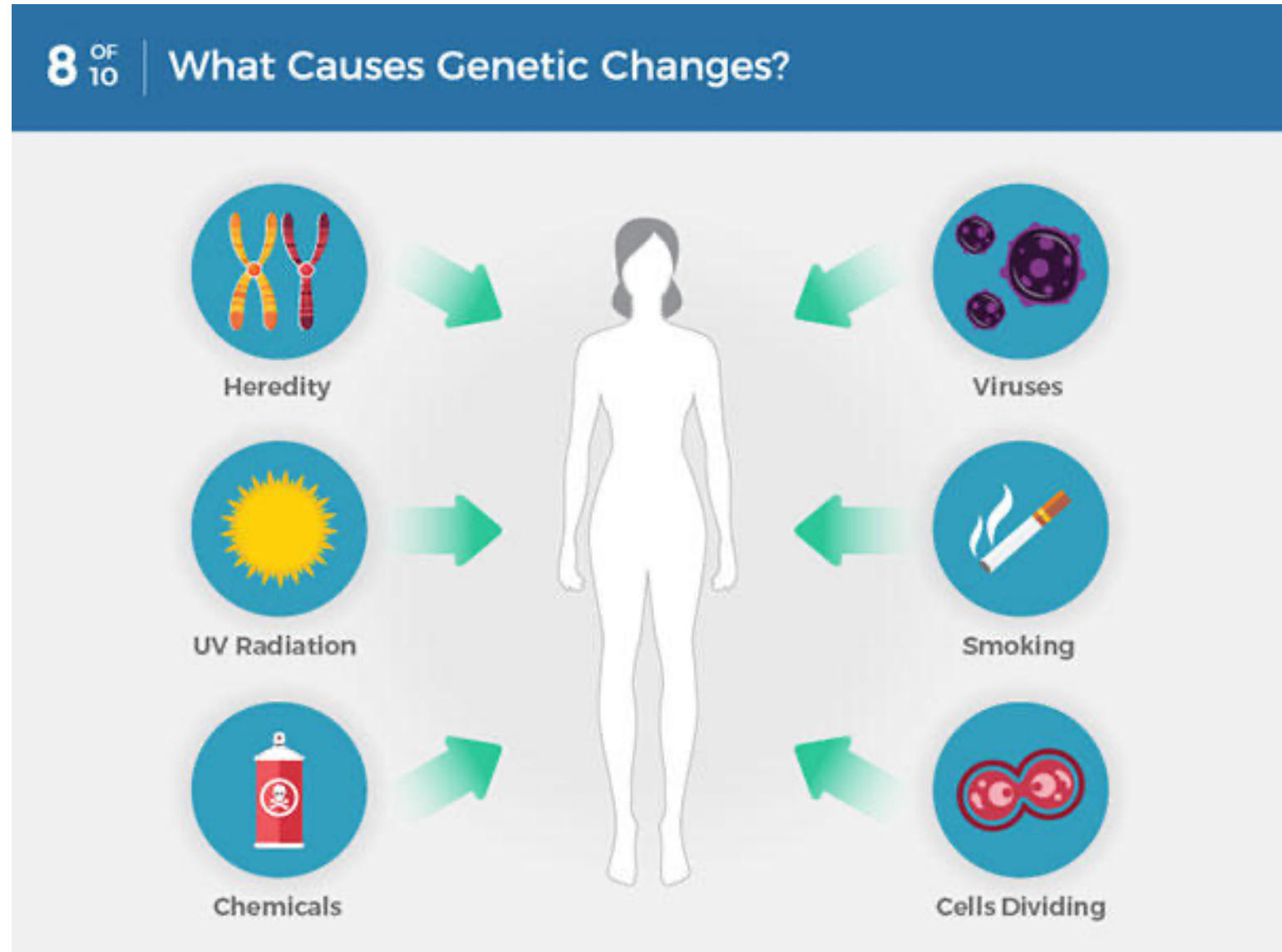
- Controlled growth
- Normal chromosomes
- Undergo apoptosis
- Contact inhibition
- Stay in organ
- Specialized cells
- Undergo maturation and senescence
- Even appearance
- Even shaped nuclei
- Organized arrangement

Cancer cells

- Uncontrolled growth
- Abnormal chromosomes
- No apoptosis
- No contact inhibition
- Ability to metastasize
- Nonspecialized cells
- Immature/undifferentiated
- Variable appearance
- Variable shaped nuclei
- Disorganized arrangement

Is cancer a genetic disease?

- Yes, cancer is a **genetic disease**.
- It is caused by changes in genes that control the way cells grow and multiply.
- DNA changes, whether caused by a random mistake or by a carcinogen, can happen throughout our lives and even in the womb.



Is cancer hereditary?

- Cancer itself can't be passed down from parents to children.
- But a genetic change that increases the risk of cancer can be passed down (inherited) if it is present in a parent's egg or sperm cells.

For example, if a parent passes a mutated BRCA1 or BRCA2 gene to their child, the child will have a much higher risk of developing breast and several other cancers.

- *That's why cancer sometimes appears to run in families. Up to 10% of all cancers may be caused by inherited genetic changes.*
- *Inheriting a cancer-related genetic change doesn't mean someone will definitely get cancer. It means that the risk of getting cancer is increased.*

Should one get genetic testing for cancer risk?

- Certain genetic tests can show if you've inherited a genetic change that increases your risk of cancer.
- This testing is usually done with a **small sample of blood**, but it can sometimes be done with **saliva, cells from inside the cheek, or skin cells**.



Genetic tests can help families with a history of breast and ovarian cancer make screening and treatment decisions

AIDS

Acquired Immunodeficiency Syndrome

- A disease caused by the human immunodeficiency virus (HIV).

What's the Difference Between HIV and AIDS?

- HIV stands for Human Immunodeficiency Virus.
It's a virus that attacks the body's immune system, weakening the defenses against infections.
- AIDS, on the other hand, stands for Acquired Immunodeficiency Syndrome.
- It's the most severe stage of HIV infection, where the immune system is so damaged that it can no longer fight off certain infections or cancers.

How HIV Infection Affects the Body



HIV/AIDS is transmitted



UNPROTECTED
SEX



DRUG ADDICTS



BLOOD
TRANSFUSION



PREGNANCY



NON-STERILE
INSTRUMENTS

HIV/AIDS is not transmitted



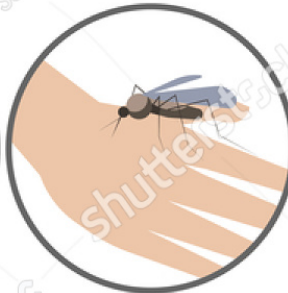
TOUCHING



THROUGH FOOD



WITH A KISS



INSECT BITES



IN THE POOL

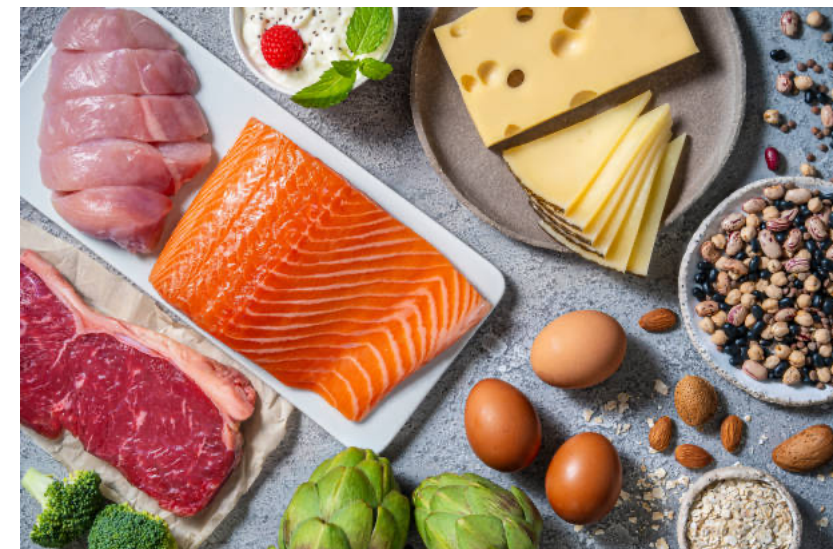
HIV Diagnosis

- Blood Test
- ELISA (Enzyme Linked Immunosorbent Assay)



What is Nutrition and why it is so important?

- The study of food, its composition, the amounts needed by the body and its effects on the body.
- At the most basic level, nutrition is about eating a regular, balanced diet.
- Good nutrition helps fuel your body. The foods you eat supply the nutrients your body needs to maintain your brain, muscle, bone, nerves, skin, blood circulation, and immune system.



- Proper nutrition also helps protect you from illness and disease, such as heart disease, diabetes, cancer, and osteoporosis.
- Nutrients are chemical substances required by the body to sustain basic functions and are optimally obtained by eating a balanced diet.

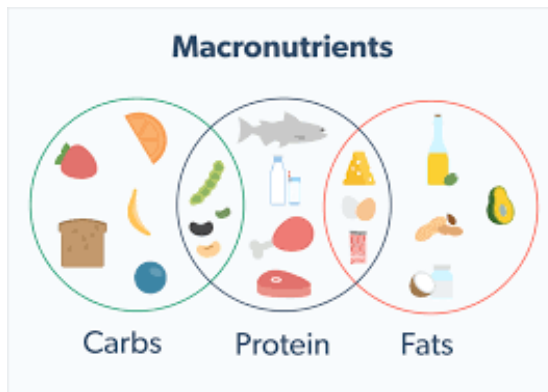
Six major classes of nutrients

- Carbohydrates
- Lipids/Fats
- Proteins
- Vitamins
- Minerals
- Water (*does not yield energy*)

Major classes of nutrients in food

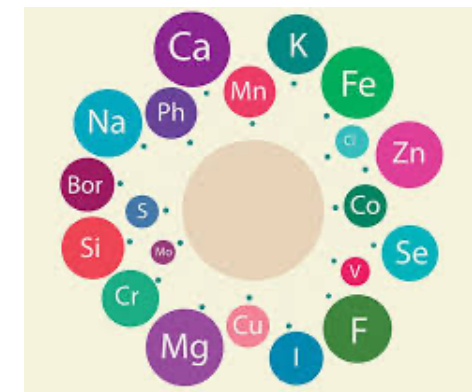
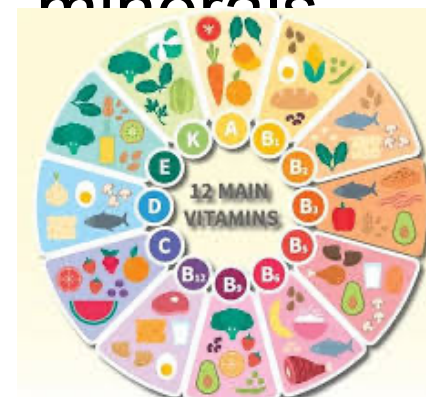
Macronutrients

- *Includes: Carbohydrates, Protein, and Fat (produce energy)*
- supply energy (in the form of calories)
- serve as the building blocks for muscles and tissues



Micronutrients

- *Includes: Vitamins and Minerals (plays a role in metabolism)*
- Divided into four categories: water-soluble vitamins, fat-soluble vitamins, microminerals, and trace minerals



Vitamins

(are **Organic** micronutrients)

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graph TD; A["Vitamins  
(are Organic micronutrients)"] --> B["Water-soluble"]; A --> C["Fat-soluble"]; B --> D["Vitamins B1, B2,  
B3, B5, B6, B7,  
B9, B12, and C"]; C --> E["Vitamins A, E,  
D, and K."];
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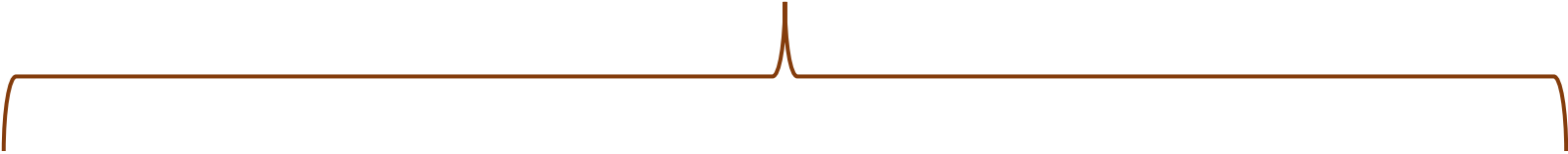
Water-soluble

Vitamins B1, B2,
B3, B5, B6, B7,
B9, B12, and C

Fat-soluble

Vitamins A, E,
D, and K.

Minerals (are **Inorganic** micronutrients)



Macrominerals

- Required in amounts greater than 100 mg per day
- Ex: Calcium, phosphorous, magnesium, sodium, potassium, and chloride.

Microminerals

- Required in amounts less than 100 mg per day
- Ex: Iron, copper, zinc, selenium, and iodine.

Carbohydrates:

- One of the essential macronutrients that are the **primary source of energy** for humans; 1 gram of carbohydrate contains 4 kcal of energy
- Carbohydrates are present in plant-based foods like **grains, fruits, vegetables, and milk.**
- Carbohydrates are ingested in the form of simple carbohydrates, like monosaccharides and disaccharides, or complex carbohydrates, like oligosaccharides and polysaccharides.
- **Glucose (blood sugar)** is the **primary** form to which carbohydrates become **metabolized** in humans.
- **Lactose** is a carbohydrate **found in milk**, and **sucrose** is basic **table sugar**.
- **Fibers** are plant carbohydrates (polysaccharides) like pectin and cellulose found in whole grains, fruits, vegetables, and legumes but **are not digestible by humans.**
- However, they **play a major role in gut health** and function and **can be digested by microbiota in the large intestine.**

There are three main types of carbohydrates

Sugars:

- They are also called **simple carbohydrates** because they are in the most basic form.
- They can be added to foods, such as the sugar in candy, desserts, processed foods, and regular soda.
- They also include the kinds of sugar that are found naturally in fruits, vegetables, and milk.

Starches:

- They are **complex carbohydrates**, which are made of lots of simple sugars strung together.
- Human body needs to break starches down into sugars to use them for energy.
- Starches include bread, cereal, and pasta.
- Also include certain vegetables, like potatoes, peas, and corn.

Fiber:

- It is also a complex carbohydrate.
- Human body cannot break down most fibers, so eating foods with fiber can help you feel full and make you less likely to overeat.
- Diets high in fiber have other health benefits. They may help prevent stomach or intestinal problems, such as constipation.
- They may also help lower cholesterol and blood sugar.
- Fiber is found in many foods that come from plants, including fruits, vegetables, nuts, seeds, beans, and whole grains.

Which foods have carbohydrates?

- Common foods with carbohydrates include:

Grains, such as bread, noodles, pasta, crackers, cereals, and rice

Fruits, such as apples, bananas, berries, mangoes, melons, and oranges

Dairy products, such as milk and yogurt

Legumes, including dried beans, lentils, and peas

Snack foods and sweets, such as cakes, cookies, candy, and other desserts

Juices, regular sodas, fruit drinks, sports drinks, energy drinks that contain sugar

Starchy vegetables, such as potatoes, corn, and peas



Which types of carbohydrates should we eat?

- When eating grains, choose mostly whole grains and **not refined grains**:
 - Whole grains are foods like whole-wheat bread, brown rice, whole cornmeal, and oatmeal.
 - They offer lots of nutrients that your body needs, like vitamins, minerals, and fiber.
 - Refined grains are foods that have had some parts of the grains removed. This also removes some of the nutrients that are good for your health.
 - Eat foods with lots of fiber.
-
- Try to **avoid** foods that have a **lot of added sugar**.
 - These foods can have many calories but not much nutrition.
 - Eating too much added sugar raises the blood sugar and can make gain weight.



How many carbohydrates one should eat?

- There is **no one-size-fits-all** amount of carbohydrates that people should eat. This amount can vary, depending on factors such as your age, sex, health, and whether or not one is trying to lose or gain weight.
- On average, people should get 45 to 65% of their calories from carbohydrates every day.
- On the Nutrition Facts labels, the Daily Value for **total carbohydrates is 275 g per day.**
- This is based on a 2,000-calorie daily diet.
- Daily Value may be higher or lower depending on your calorie needs and health.

Is it safe to eat a low-carb diet?

- Some people go on a **low-carb** diet to try to lose weight.
- This usually means eating between **25 g and 150 g** of carbs each day.
- This kind of diet can be safe, but one should talk to the health care provider before starting it.
- One problem with low-carb diets is that they can limit the amount of fiber you get each day.
- They can also be hard to stay on for the long term.

Proteins

- Protein is in every cell in the body.
- Our bodies need protein from the foods we eat to build and maintain bones, muscles and skin.
- Proteins source: meat, dairy products, nuts, and certain grains and beans.
- Proteins from meat and other animal products are **complete proteins**. This means they supply all of the amino acids the body can't make on its own.



- Most plant proteins are incomplete.
- One should eat different types of plant proteins every day to get all of the amino acids your body needs.
- It is important to get enough dietary protein, Because human body doesn't store it the way it stores fats or carbohydrates.
- Protein requirement depends on your age, sex, health, and level of physical activity.
- 1 gram of protein contains 4 kcal of energy.
- The recommended protein intake is 0.8 to 1 gram per kilogram of body weight per days

For healthy children ages

1 to 3: 5 to 20%,

4 to 18: 10 to 30%,

Adults: 10 to 35%

of daily energy intake should come from protein, respectively.

Lipids/Fats

- Fat is a type of nutrient.
- Fats gives energy and help the body absorb vitamins.
- Lipids are **essential macronutrients** that are the **main source of stored energy** in the body, contribute to cellular structure and function, regulate temperature, and protect body organs.
- Lipids are found in fats, oils, meats, dairy, and plants and consumed mostly in the form of triglycerides
- One gram of fat contains 9 kcal of energy.



- But not all fats are the same. One should try to **avoid:**
Saturated fats such as butter, solid shortening, and lard
Trans fats.
- These are found in vegetable shortenings, some margarines, crackers, cookies, snack foods.

For healthy children ages:

1 to 3: 30 to 40%,

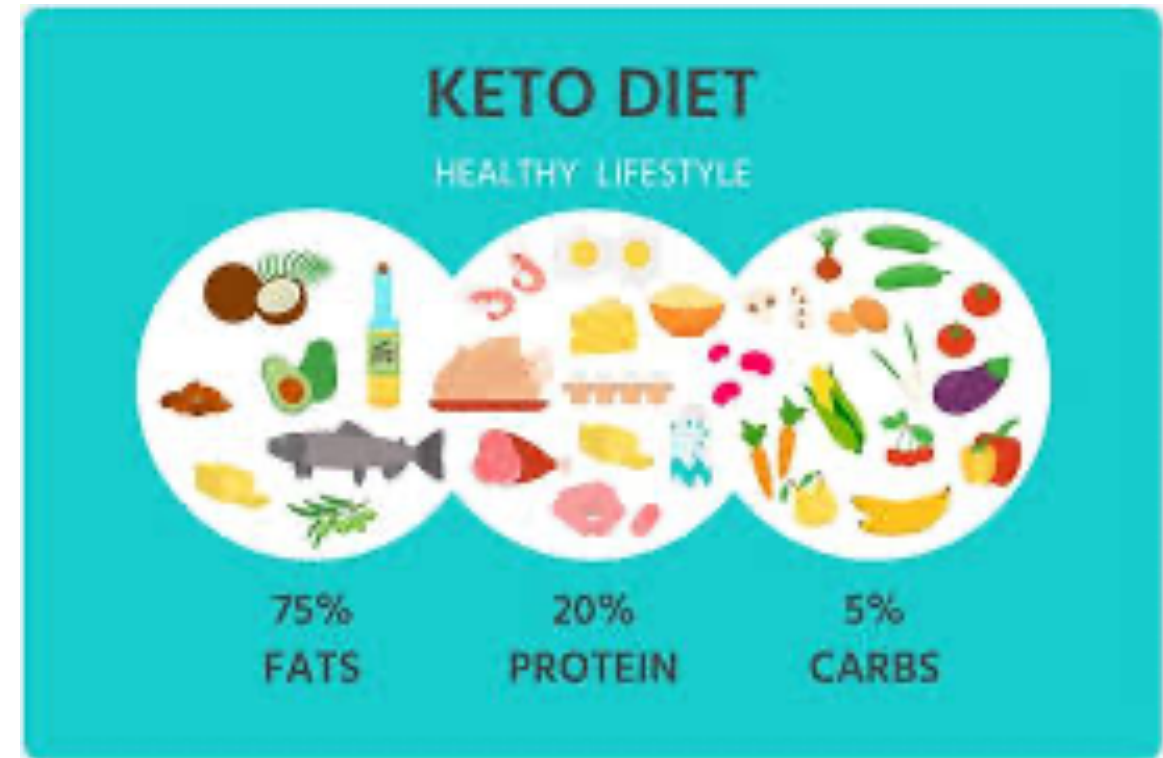
4 to 18: 25 to 15%,

Adults: 20 to 35%

of daily energy intake should come from fat

Keto Diet

- The ketogenic diet is a very low carb, high fat diet
- It involves drastically reducing carbohydrate intake and replacing it with fat. This reduction in carbs puts your body into a metabolic state called [ketosis](#).



Ketosis is a metabolic process. When the body does not have enough glucose for energy, it burns stored fats instead. This results in a buildup of acids called ketones.

- In keto diet; body becomes incredibly efficient at [burning fat](#) for energy.
- It also turns fat into [ketones](#) in the liver, which can supply energy for the brain

Is the keto diet healthy?

- Keto diet could reduce a person's risk of developing several health conditions, including metabolic syndrome, [cardiovascular disease](#), and diabetes.
- These health benefits may occur due to the loss of excess weight and the inclusion of more nutritious foods in the diet, rather than the reduction in carbohydrates.

Vitamin B1 (Thiamin)

- Thiamin, or vitamin B1, is an essential water-soluble vitamin
- Acts as a coenzyme in carbohydrate and branched-chain amino acid metabolism.
- Thiamin is in food sources such as enriched and whole grains, legumes, and pork.
- The RDA (Recommended Dietary Allowance) of thiamin for adults is 1.1 mg/day for women and 1.2 mg/day for men.



Vitamin B2 (Riboflavin)

- Riboflavin, or vitamin B2, is an essential water-soluble vitamin
- Acts as a coenzyme in redox reactions
- Riboflavin is present in food sources such as enriched and whole grains, milk and dairy products, leafy vegetables, and beef.
- The RDA of riboflavin for adults is 1.1 mg/day for women and 1.3 mg/day for men.

Vitamin B3 (Niacin)

- Niacin, or vitamin B3, is an essential water-soluble vitamin
- Niacin is present in enriched and whole grains and high protein foods like meat, milk, and eggs.
- The RDA of niacin for adults is 14 mg/day of NEs (niacin equivalents) for women and 16 mg/day of NEs for men

Vitamin B12 (Cobalamin)

- Vitamin B12, or cobalamin, is an essential **water-soluble vitamin**. Vitamin B12 is **only present in animal products** because it is a product of bacteria synthesis.
- Many foods are also fortified with synthetic vitamin B12.
- The RDA of vitamin B12 for adults is **2.4 mg/day**.
- It is recommended for older adults to meet their RDA with fortified foods or supplements because many are unable to absorb naturally occurring vitamin B12.

Vitamin C (Ascorbic Acid)

- Vitamin C, or ascorbic acid, is an essential **water-soluble** vitamin
- Acts as a reducing agent in enzymatic reactions and nonenzymatically as a soluble antioxidant.
- Vitamin C is found primarily in **fruits and vegetables, except for animal organs like the liver and kidneys.**
- The RDA of vitamin C for adult women and men is 75 mg/day and 90 mg/day, respectively.

